

KAUSHALRAJSINH PUWAR

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EDUCATION

International Institute of Information Technology Bangalore (IIIT-B)

Jul 2023 – Present

Integrated M.Tech in Computer Science and Engineering

CGPA: 2.89/4.00

EXPERIENCE

OpenG2P

May 2026 – Present

Machine Learning Intern

- Built an **LLM-assisted pipeline** converting natural-language queries into schema-constrained SQL with deterministic validation, retry logic, and multi-source routing.
- Evolved the static pipeline into a **live-loop agent** with sequential tool calling, mid-investigation clarification, and adaptive strategy based on intermediate results.
- Enabled **cross-source analysis** across 4 database systems with structured output contracts, logging, and confidence scoring, reducing manual triage effort.

PROJECTS

AutoDocLM | Python, Ollama, OpenAI SDK, ChromaDB, NetworkX, MkDocs 🔄 [code]

Mar 2026 – Apr 2026

- Developed an **LLM-powered code understanding system** that transforms large repositories into structured documentation, architecture summaries, and Mermaid dependency diagrams.
- Built a static analysis pipeline using **AST parsing, graph analytics, and PageRank centrality** to model cross-file dependencies and prioritize high-impact code components.
- Integrated **retrieval-augmented generation (RAG)** with local embeddings, ChromaDB, and external inference APIs to generate architecture-aware technical documentation at repository scale.

Thermal Plant Control Environment | Python, OpenEnv, Pytorch, Docker, HF Spaces 🔄 [code]

Mar 2026 – Apr 2026

- Built an **OpenEnv-compatible RL benchmark** evaluating LLM reasoning in safety-critical control; simulated a thermal-hydraulic plant with actuator lag, thermal inertia, and hardware degradation.
- Designed a **composite reward function** across tracking error, safety violations, and oscillation with catastrophic failure penalties and an anti-coasting evaluation window activated post-disturbance.
- Benchmarked Llama-3.3-70B vs. rule-based baseline across **4 escalating tasks**; LLM achieved **54% reward improvement** in Fault Recovery under hardware degradation.

Adaptive Bridge | ROS 2, Python, Fast DDS, Cyclone DDS, Docker 🔄 [code]

Jan 2026 – Mar 2026

- Built a **proxy-based middleware** that isolates critical and noncritical ROS 2 subscribers via independent DDS writers, eliminating slow-subscriber backpressure.
- Reduced critical subscriber tail latency **from 15 s to 1.55 ms at p95 (~9,700x)** and preserved publisher throughput at **30 Hz** under Gilbert-Elliott bursty loss.
- Validated across Fast DDS and Cyclone DDS using a reproducible Docker-based testbed; wrote an IEEE RA-L manuscript targeting mixed-criticality robotics networking.

Signal Cluster Classification | Python, Scikit-learn, PyTorch, Jupyter 🔄 [code]

Oct 2025 – Nov 2025

- Benchmarked **8+ classifiers** - GMM, K-Means, SVM (Linear & RBF), MLP, Random Forest, GNB, and two GNN variants - with model-specific preprocessing pipelines for each approach.
- Conducted structured EDA via **PCA, correlation heatmaps**, and class-means visualizations; applied probability calibration on MLP outputs to improve confidence reliability.
- Achieved **Macro F1 of 0.989** via **ensemble stacking**, outperforming all individual baselines including GNN architectures.

TECHNICAL SKILLS

- Programming Languages:** Python, C/C++, Go, Rust, JS/TS, Java, HTML/CSS, PostgreSQL
- Libraries & Frameworks:** PyTorch, NumPy, Hugging Face, Gradio, React, Spring Boot, Node.js, Tailwind, Matplotlib, ROS 2
- Tools & Platforms:** MongoDB, Docker, AWS, Bedrock, Kubernetes, Jenkins, Anaconda, Jupyter Notebook, Android Studio, Linux

EXTRACURRICULARS

AWS Student Builders Group | Core Member

Jun 2026 – Present

- Part of the core organizing team responsible for supporting AWS-focused workshops, technical events, and student community initiatives.

Infin8 Sponsorship Team | Sponsorship Coordinator

Nov 2023 – Feb 2024

- Contributed as the Sponsorship Coordinator for the annual cultural festival Infin8 and secured sponsorships for the event